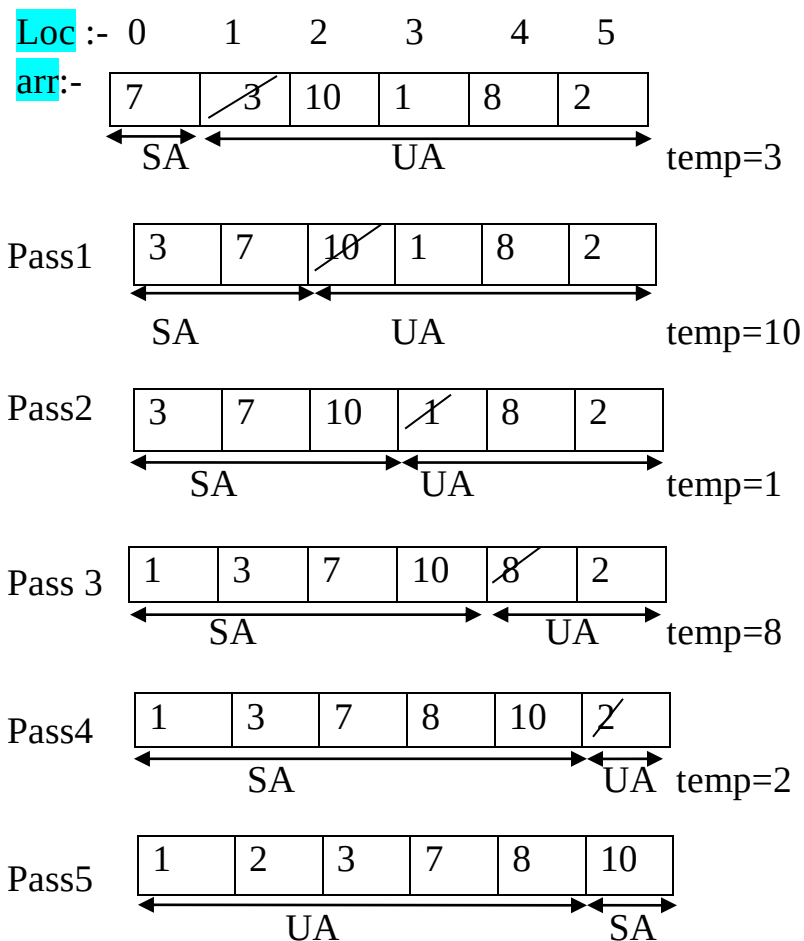


INSERTION SORT ALGORITHM:-

Insertion sort algorithm divides array into two sub lists one is sorted list and other is unsorted list. The idea behind the insertion sort is that first take one element, iterate it through the sorted array and keep on doing till the whole array is sorted.

Insertion sort works similar to the sorting of playing cards in hands. It is assumed that the first card is already sorted in the card game, and then we select an unsorted card. If the selected unsorted card is greater than the first card, it will be placed at the right side; otherwise, it will be placed at the left side. Similarly, all unsorted cards are taken and put in their exact place. Time complexity of algorithm is $O(n^2)$.

Example :- (For N=6)



```
int insertionsort (int arr[ ]) {  
    for( i=1 ; i < n; i++) {  
        temp=a[ i ];  
        j= i -1;  
        while(j >=0 && a[j] > temp) {  
            a [ j+1] = a[ j ];  
            j--; }  
        a [ j+1] = temp;  
    }  
}
```

Int insertion sort (int a[])	Time Complexity
{	
int i;	1
int j;	1
int temp;	1
{	
for(i=0; i < n; i++)	$1+N-1+1+N-1= 2N$
{	
temp =a[i]	$2(N-1)$
j = i-1;	$2(N-1)$
While(j >= 0 && a[j] > temp)	$3x$ (assume $N-1=x$)
{	
A[j+1] = a[j];	$2x$
j--;	x
}	
A[j+1] = temp;	$2(N-1)$
}	
}	
	$T(N)= 1+1+1+2N+2N-2+2N-2+3x+2x+x+2N-2$
	$6x+8N+1$
	$\therefore X= 1 + 2+3 +\dots\dots(N-1)$ in Arithmetic progression
	Considering AP $1+2+3+\dots\dots k = K(K+1)/2$
	$= 6N^2/2-6N/2+8N+1$
	$= 3N^2- 3N+8N+1$
	Considering big O notation and ignoring constant we consider the term of higher growth.
	$T(N) = O(N^2)$.

Assume that you are given **four number-display-cards** (i.e., the cards having some number written on each of them). Assume that you cannot have a look at more than two cards at a time to reveal the numbers written on the cards. How many comparisons you would require to have them arranged in ascending order according to the numbers written on the cards?

Show Working steps how?

(Remember, you can only compare the numbers written on the two cards if both of them are in front of you.)

Values for 4 Play Cards are

5 9 3 1

Step 1 Compare first two play cards **5** with **9** and get the highest number **9**.

step 2 Compare the highest number from step1 with another play card **9** and **3** and get the highest number **9**.

Step 3 Compare the highest number from step 2 with another play card, **9** and **1** and get the highest number **9**.

Remaining Cards are

5 3 1

Step 4 Compare first two play cards **5** and **3** and get the highest number **5**.

Step 5 compare highest number from **step 1** with another play card **5,1** and get highest number **5**.

Remaining Cards are

3 1

Step 6 Compare the remaining cards are **3** and **1** and get the highest no **3**.

Conclusion:

Finally we have total 6 no. of comparisons.